

SANTHOSH BABU

AI Engineer | Generative AI | Agentic AI | LLMs | RAG | Python

santhoshbaskaran12@gmail.com | +91-9080501853 | Chennai, TamilNadu- 600034

PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience building and deploying Generative AI and Agentic AI solutions across banking, retail, and supply chain domains. Self-transitioned into AI engineering through intensive hands-on project work, bringing deep expertise in designing RAG pipelines, multi-agent systems, and LLM-powered applications using Python, LangChain, LangGraph, and vector databases. Comfortable working across the full AI development lifecycle — from prompt engineering and model fine-tuning to API deployment on AWS and Azure. Known for translating business problems into practical AI solutions that reduce manual effort, improve accuracy, and scale reliably in production.

CORE COMPETENCIES

Programming: Python, SQL

Generative AI: GPT-4, LLaMA 2/3, Mistral, LangChain, LangGraph, LoRA/QLoRA Fine-Tuning

RAG & Vector DB: RAG Pipelines, FAISS, Pinecone, ChromaDB, HuggingFace, Semantic Search

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, NLP, Text Classification, MLflow

Agentic AI: Multi-Agent Systems, LangGraph Agents, Tool Use, Autonomous Workflows, Human-in-the-Loop

Cloud & Deployment: AWS (S3, Lambda, Step Functions), Azure OpenAI, FastAPI, Streamlit, REST APIs

Databases: PostgreSQL, MySQL, ChromaDB, Pinecone

Tools: Git, Pandas, NumPy, Matplotlib, NLTK, Jupyter

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Centaur Billing IT Solutions

April 2024 – Present

- Designed and deployed end-to-end Generative AI applications using GPT-4, LLaMA, and Mistral — covering document intelligence, NLP automation, and financial decision-support across multiple business units.
- Built production RAG pipelines using Pinecone and ChromaDB with HuggingFace embeddings, achieving over 90% retrieval accuracy; fine-tuned LLaMA models using LoRA/QLoRA to reduce hallucinations in domain-specific responses.
- Developed NLP pipelines for entity recognition, text classification, and summarisation using PyTorch and TensorFlow — reducing manual document review effort by 35%.
- Deployed AI services on AWS using Docker and FastAPI; integrated with existing systems via REST APIs with reliable performance under production load.

KEY PROJECTS

1. FinCopilot — Autonomous AI Copilot for Banking Operations [Banking]

Stack: Python, GPT-4, LangChain, LangGraph, Pinecone, HuggingFace, FastAPI, AWSLambda

- Built a multi-agent AI copilot using LangGraph where each agent handled a specific task — query resolution, anomaly triage, and regulatory report generation — replacing 3 manual banking workflows.
- Implemented a hybrid RAG + Text-to-SQL system using HuggingFace embeddings with semantic chunking, allowing business users to query over 50M+ records in natural language at sub-2-second response time; achieved 92% first-contact resolution rate.

2. DemandSense — Retail Demand Forecasting & Inventory Intelligence [Retail]

Stack: Python, GPT-4, LangChain, XGBoost, PyTorch, Scikit-learn, MLflow, AWS S3, Streamlit

- Built a demand forecasting system using XGBoost and PyTorch LSTM models trained on historical sales data, predicting stock requirements across 500+ SKUs with 87% accuracy; tracked experiments and model versions using MLflow.
- Added an LLM layer using LangChain to convert model predictions into plain-language reorder alerts and supplier briefings automatically — reducing overstock incidents by 32% and stockout events by 41%.

3. ProcureIQ — Agentic Procurement & Vendor Risk Intelligence Platform [Supply Chain]

Stack: Python, GPT-4, LangGraph, LangChain, ChromaDB, PostgreSQL, FastAPI, Streamlit

- Built a multi-agent system using LangGraph where agents autonomously reviewed vendor proposals, extracted key contract terms, flagged anomalies against compliance rules, and produced risk-scored evaluation summaries — cutting procurement review time by 60%.
- Implemented a RAG pipeline over 10K+ vendor contracts and policy documents using ChromaDB, enabling teams to query procurement records in natural language with over 88% retrieval accuracy and real-time risk alerts.

KEY ACHIEVEMENTS

- Achieved 92% first-contact resolution and sub-2-second query latency in banking AI copilot handling 50M+ records.
- Delivered 87% demand forecasting accuracy across 500+ SKUs — reducing overstock by 32% and stockout events by 41%.
- Improved recommendation click-through by 38% and average order value by 22% via agentic personalisation engine.
- Reduced manual document review effort by 35% through NLP pipelines built with PyTorch and TensorFlow.
- Cut procurement review time by 60% and achieved >88% RAG retrieval accuracy across 10K+ vendor contracts.
- Enabled business users to self-serve retail analytics in natural language — eliminating dependency on analyst teams.

EDUCATION

Bachelor of engineering - Mechanical Engineering - CGPA:8.2

Hindusthan college of engineering and technology, Coimbatore | Graduated 2024

CERTIFICATIONS

- Postgraduate Program In Data Science and Artificial Intelligence, Coimbatore (2024)

SOFT SKILLS

Analytical Thinking | Problem Solving | Stakeholder Management | Cross-functional Collaboration | Attention to Detail | Adaptability | Continuous Learning